

Experiment 1: Verification of Ohm's Law

1. Objectives:

- To confirm Ohm's law by studying the relationship of voltage, current and resistance.
- To make voltage and current measurements using a DMM in a resistive circuit to verify Ohm's law.

2. Backgrounds:

Ohm's Law: Ohm's law states that electrical current in a resistive circuit is *directly proportional to the applied voltage* and *inversely proportional to its resistance*, provided all physical conditions and temperatures remain constant.

$$I = \frac{V}{R} \quad \text{in amperes (A)}$$

Where V is the applied voltage in volts (V).
 R is the resistance in ohms (Ω).
 I is the current in amperes (A).

The larger the applied voltage is, the larger the current becomes. The larger the resistance is, the smaller the current becomes.

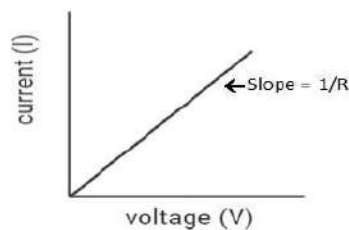
Conditions of Ohm's Law:

- A conducting device obeys Ohm's law when the resistance of the device is independent of the magnitude and polarity of the applied potential difference.
- Ohm's law is not applicable for unilateral electrical elements like diodes and transistors as they allow the current to flow through in one direction only.

Graphical Representation of Ohm's Law: Ohm's law can be represented as a straight line in x-y coordinates as the current and the voltage have a linear relationship.

$$I = \frac{V}{R} = \frac{1}{R} \cdot V$$

where, $\frac{1}{R}$ represents the slope of the line with I on the vertical axis and V on the horizontal axis.



Voltage- Current Characteristics- Ohm's Law Graph

3. Procedures and Observations:

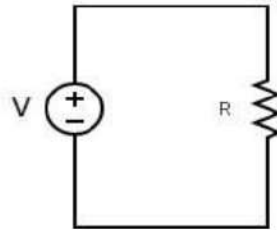


Figure 1

1. Pick $2.2k\Omega$, $4.7k\Omega$, $10k\Omega$, from your laboratory kit as R_1 , R_2 and R_3 .
2. Measure the resistance using the DMM and record the measured value in Table 1.
3. Connect the circuit shown in Figure 1 using R to start. Set DMM to measure voltage and current simultaneously or separately.

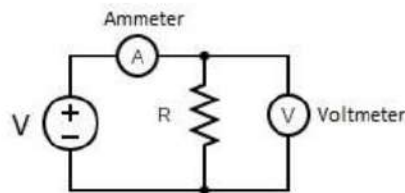


Figure 2

4. **Before turning on the DC power supply, have your circuit checked by the instructor.**
5. With the voltage control knob set to the minimum (fully CCW), turn the DC power supply on and set the voltage control to each voltage value shown in Table 1, beginning with 1 V.
6. For each voltage setting, measure and record the actual voltage and current values in Table 1 (Page no. 4).
7. Turn the power supply off.
8. Calculate the theoretical current " $I_{calculated}$ " values using the nominal resistance values and the nominal voltage values " $V_{Nominal}$ ".
9. Compare the calculated currents with the measured, and if satisfied, proceed to the next step.
10. Repeat the above procedures from step 2 through 9, using the other two resistors.

Lab Report:

Date:	
Name of the Students and IDs	(1)
	(2)
	(3)

Data Table:

	R ₁			R ₂			R ₃		
	Nominal R: $2.2\text{ k}\Omega$			Nominal R: $4.7\text{ k}\Omega$			Nominal R: $10\text{ k}\Omega$		
	Measured R:			Measured R:			Measured R:		
	Measured Values		$I_{\text{Calculated}}$	Measured Values		$I_{\text{Calculated}}$	Measured Values		$I_{\text{Calculated}}$
V_{Nominal}	V_{Measured}	I_{Measured}		V_{Measured}	I_{Measured}		V_{Measured}	I_{Measured}	
1 V									
2 V									
5 V									
10 V									
15 V									
20 V									
25V									

Table 1: Voltage and Current Measurement and Calculation

Tasks and Questions:

#1: Comment on the measured current compared to calculated currents.

#2: Use the data obtained in Table 1 to plot I_{measured} vs. V_{measured} graph.

#3: What does the inverse of the slope of your graphs represent? Illustrate with an example. [Hint: Find slopes from the graph].

#4: Using your graph, estimate the current that would flow through the resistors at $V = 12$ volts and compare it with the calculated value ($12 \text{ V} / R_{\text{Nominal}}$). Calculate the error.

For $E = 12 \text{ V}$

Nominal Resistance	Estimated Current from Graph	Calculated Current ($12 \text{ V} / R_{\text{Nominal}}$)	% of Error
$R_1 =$			
$R_2 =$			
$R_3 =$			

Table 2: Current Estimation and Calculation at $V = 12 \text{ V}$

Results:

Discussion: